

Group: ZJU-China 2026 WW2

Selected Part Name: To be assigned

Part Type: coding

Short Description: Peptide introducing N-MYC aggregation

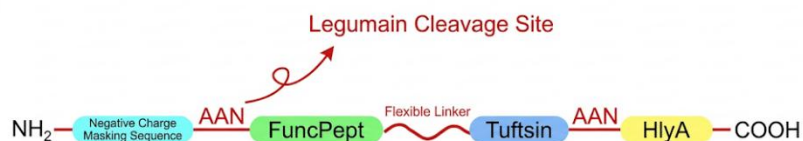
Long Description: This basic part is a fully optimized coding DNA sequence (CDS) that expresses an engineered peptide to induce targeted aggregation of N-MYC proteins. At the genetic level, this sequence has been extensively codon-optimized for *E. coli* chassis, while functionally, its translated product is designed to induce irreversible aggregation of oncogenic transcription factor N-MYC in Small Cell Lung Cancer (SCLC).

Sequence:

ATGGCGGAGGACGAAGATGAAGATGCGGCCAATAAAACCACATTTACGATC
ACCAAGGGCACCACCTTTTACGATCACTAAGGGTGGCGGTGGATCGACCAA
ACCTCGTGCGGCAAACTTGGCATAACGGTAGTCAAGGCGATCTGAACCCGCT
TATTAATGAGATCAGCAAAATTATTTCTGCCGCAGGTTTCCTTTGACGTTAAG
GAAGAGCGCACCGCTGCGTCTTTGTTGCAACTGTCCGGTTAA (246bp)

Amino Acid Sequence:

MAEDEDEDAANKTTFTITKGTTFITKGGGGSTKPRAANLAYGSQGDLNPLIN
EISKIISAAGSFDVKEERTAASLLQLSG



Source:

Acidic Modulator (aa 3-8): A highly acidic stretch (EDEDED) designed to modulate the overall charge, potentially assisting in early folding or preventing premature self-aggregation before reaching the target.

Targeting & Aggregation Domains (aa 12-26): Two tandem repeats of specific β -sheet-prone binding motifs (NKTTFITK / KGTTFTITK). This region is de novo synthetic sequence to recognize and co-aggregate with the intrinsically disordered regions of N-MYC.

Flexible Linker (aa 28-32): A Gly-Ser rich linker (GGGGS) at the C-terminus provides structural flexibility, ensuring the targeting domains are fully exposed for target engagement.

Immunomodulatory Tag(33-36): The C-terminal tetrapeptide TKPR (Tuftsin) is naturally derived from the Fc-domain of human Immunoglobulin G (IgG) (residues 289-292).

Design Considerations: `

To ensure high compatibility, genetic stability, and robust continuous secretion in our *E. coli* chassis, several crucial engineering optimizations were applied to the raw amino acid sequence during its back-translation to DNA:

RFC[10] Standard Compatibility: The coding sequence was strictly optimized in silico to eliminate all illegal restriction sites required by the BioBrick RFC[10] standard (EcoRI, XbaI, SpeI, PstI, and NotI). This ensures the part is "scarless-assembly ready" for standardized downstream cloning.

Elimination of Direct Repeats via Codon Degeneracy: Our Pept-in design features a tandem aggregation core (KTTFTITK and GTTFTITK) to thermodynamically enhance β -sheet nucleation. In a naive reverse-translation, this tandem repeat would result in long direct nucleotide repeats, significantly increasing the risk of homologous recombination and sequence deletion in *E. coli*. We strategically utilized codon degeneracy (e.g., coding the first Threonine-rich segment as ACCACATTTACGATCACC and the second as ACCACTTTTACGATCACT) to completely break the DNA-level homology while maintaining the exact functional amino acid output.

Host-Specific Translational Optimization: The sequence was optimized specifically for *E. coli* expression machinery. For instance, the critical Arginine (R) residue in the Tuftsin targeting motif (TKPR) was encoded using the highly preferred CGT codon, avoiding the rare AGG/AGA codons that commonly cause ribosomal stalling and translational frameshifts in prokaryotes. Furthermore, a canonical ATG start codon was introduced to ensure robust cytosolic translation initiation.

T1SS Structural Constraint Adherence: To enable continuous extracellular pumping

via the Type I Secretion System (T1SS), the 42-amino-acid HlyA signal peptide must be located at the absolute C-terminus. We engineered the sequence to ensure the native TAA stop codon immediately follows the final Glycine (G) of the HlyA tag. This strictly prevents any unintended C-terminal extensions (such as standard plasmid-derived His-tags) which would sterically abolish HlyB/HlyD recognition and secretion channel assembly.

Design

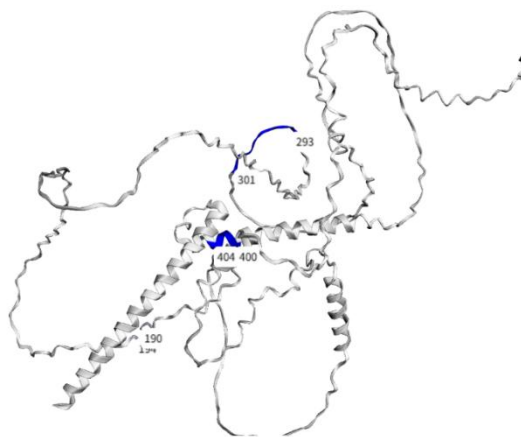
Therapeutic Peptide designed to induce targeted aggregation of oncogenic N-MYC proteins in SCLC, featuring an N-terminal signal peptide.

Biology and Usage

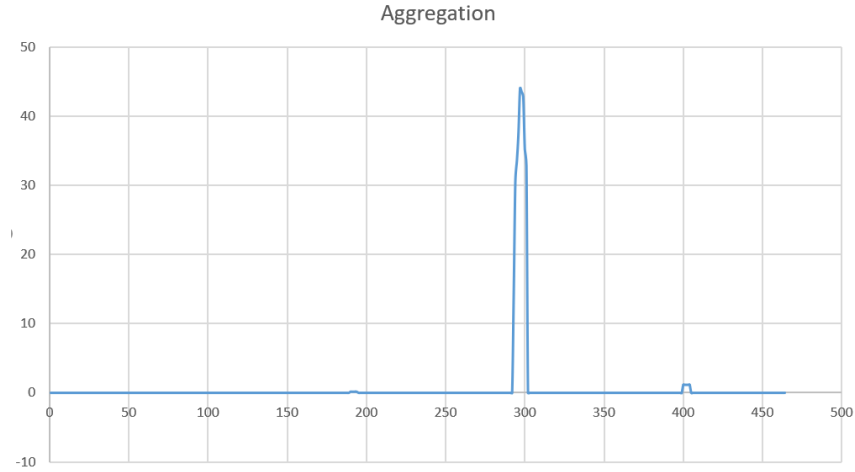
This basic part encodes a engineered peptide targeted aggregation inducer, specifically designed for therapeutic applications in Small Cell Lung Cancer (SCLC). The primary mechanism of this sequence is to target critical oncogenic transcription factors, specifically N-MYC, and induce their irreversible aggregation, thereby neutralizing their oncogenic functions and promoting cancer cell apoptosis. Intracellularly, it targets critical oncogenic transcription factor N-MYC. Extracellularly, it acts as an immunotherapeutic adjuvant in the tumor microenvironment (TME).

Proof

We utilized TANGO software to analyze the aggregation-prone regions (APRs) of the N-MYC protein, integrating these findings with AlphaFold-generated structural models as illustrated below (Blue regions indicate the APRs).



Structure of N-MYC predicted by AlphaFold



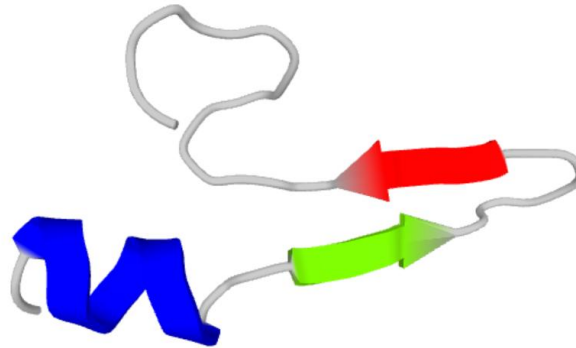
TANGO score of N-MYC per residue

Based on the identified APRs of N-MYC, several functional peptides (FuncPepts) were designed and further evaluated using the WALTZ scoring algorithm. This screening process allowed us to isolate the candidate peptide with the highest score.

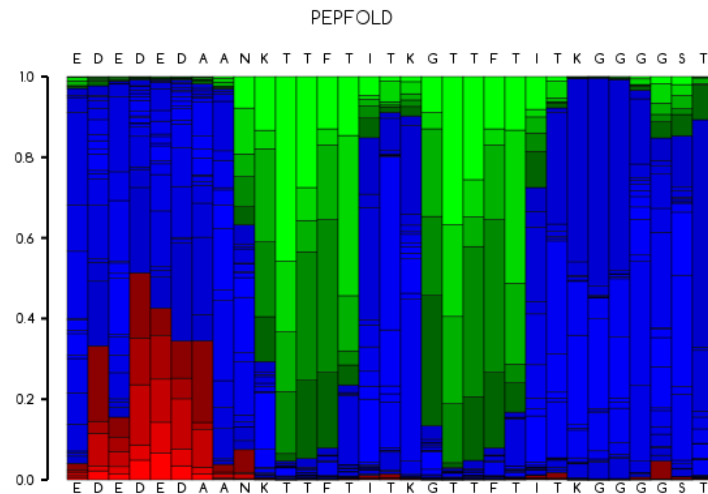
Sequence	Average Score
TTFTITKGTTFTITK	93.31104
TTFTITGKTTFTITK	93.31104
KAVTTFTIGKTTFTITK	91.44206
TTFTITKGAVTTFTIK	90.2592
TTFTITGKAVTTFTIK	90.15468
KAVTTFTIKGTTFTITK	90.06492
TTFTITKGAVTTFTK	89.47603
TTFTITGKAVTTFTK	89.36455
KAVTTFTIGKAVTTFTIK	88.81457
TTFTITGKTTFTITVK	88.04348
TTFTITKGTTFTITVK	88.04348
KAVTTFTIGKAVTTFTK	88.03856
TTFTITKGTTFTITVK	87.80379
KAVTTFTIKGAVTTFTIK	87.79264
TTFTITGKTFTITVK	87.60312
KAVTTFTIKGAVTTFTK	86.95652
KAVTTFTIGKTTFTITVK	86.86362
KAVTTFTIGKTFTITVK	86.48436
KAVTTFTIKGTTFTITVK	85.56299
KAVTTFTIKGTTFTITVK	85.48102
KAVTTFTKGTTFTITVK	84.44816
KAVTTFTGKTFTITVK	84.26003
TTFTITVKGTTFTITVK	82.88043
KTFTITVKGTTFTITVK	82.79682
TTFTITVGKTFTITVK	82.69231
KTFTITVGKTFTITVK	82.608

WALZE scores of peptides

Following the structural refinement of the lead peptide, we performed structural prediction using PEP-FOLD (results shown below).



Structure of the peptide



Predicted Local Structure

Experimental validation of the protein expression within cell models is currently pending.